

UNIT - IV

METHODS OF TEACHING

Methods of teaching

A method is "a scientific way of presenting the subject keeping in mind the psychological and Biological requirements of children."

A Biological Teacher has a variety of methods and techniques available for the use in teaching Biological Science. The selection of suitable method depends upon the objectives of the lesson, nature of the content and needs of the learner. Some methods are more appropriate for teaching students as a group where as some methods are specially designed for individual instruction. A method best for one teacher and applicable for a class, may totally be a failure for another teacher to teach the some other class. So it should be left for the teacher from his wide information, which of the methods to use and when. Here we are going to discuss in detail the various methods available to teach Biological science.

4.1 Teacher-centred methods

In these methods the teacher role is to give knowledge and full control of the classroom.

4.1.1. Lecture Method

It is the oldest teaching method. It is the most dominating method and is liked by the majority of Teachers. In this method teacher present the subject matter and the students listens. Students are not participating in this method. This method is teacher centred and information

centred and in this method teacher works as a sole resource in classroom induction. Due to lack of participation, students get bored and some of them may go to sleep. Here teacher is presenting ready made knowledge and due this spoon feeding. The students loose interest and their power of reasoning and observation get no stimulus.

In this method teacher goes ahead with the subject matter at his own speed. The teacher may use blackboard at times and may also dictate notes. This teacher centred method in its extreme form does not expect any question or response from the students. It helps only intelligent students who can understand and remember the facts told by the teacher, which is just a blind exercise.

In this method, teacher prepares his lesson plan at home and delivers it out in the class. This method does not cater for releasing the teaching Biological science and is not in accordance with the principles of teaching.

Guidelines for preparing a Lecture

To make lecture method effective, the teacher has to follow certain guidelines while preparing the lecture.

- Lecture should be carefully and systematically planned as lesson plan
- Teaching objectives should be kept in mind while preparing the lecture.
- The lecture should exhibit the through knowledge of the subject matter, its organisation, development, interpretation and application.
- Lecture should be related with day to day life experience of the students.

- Student participation should be encouraged by asking questions, clarifying doubts and reviewing the key points at frequent intervals.

Merits of Lecture method

1. It is possible of handle a large number of students at a time and no laboratory, equipment, aids are required.
2. Using this method the knowledge can be imparted to the students quickly and the prescribed syllabus can be covered in a short time.
3. Teachers feel secure and satisfied.
4. Using this method it is quite easy to impart factual information, historical anecdotes and life history of eminent personalities.
5. Teacher can develop his own style of teaching.
6. In this method teacher can easily the logical sequence of the subject by planning his lecture.
7. This method gives the students as well as the teacher a sense of satisfaction and achievement.
8. Lecture method trains students to listen long duration.
It delivers a lot of information in a short amount of time.

Demerits of Lecture method

1. In this method students participation is negligible and students become passive recipients of information.
2. We are not sure if the students are concentrating and understanding the subject matter.

3. This method leads weak students develop a hatred for learning.
4. It does not take into account the previous knowledge of the students.
5. It does not cater to the individual needs and differences of the students.
6. It is the lowest retention value of all teaching techniques.
7. It is an undemocratic and authorisation method.
8. Inability to understand one essential point may make the rest of the lecture unintelligible.
9. This method does not help in developing problem solving skills.
10. This method can be used only in higher secondary level or college level.

4.1.2. Demonstration Method

Demonstration method is a method of teaching such subjects that need live guidance on the part of the teacher. Demonstration involves showing by reason, explaining or making clear by use of examples or experiments. This strategy is applied mainly in teacher education programmes. Demonstration often occurs when students have a hard time connecting theories to actual practice or when students are unable to understand application of theories. The demonstration method of teaching shows learners how to perform the task step by step so that the learner will eventually be able to complete the same task independently.

It is a visual approach to examining information, ideas or processes. It is a teaching method that allows students

to see the teacher actively engage as a learner and a model rather than merely telling them what they need to know.

Demonstration method focus to achieve psychomotor and cognitive objectives. It is given in three successive steps.

Steps of Demonstration method

Demonstration plays an important role in teaching science. Curiosity and keen observing ability are instilled among the students

1. Planning
2. Introducing the lesson
3. Experimentation

1. Planning

Proper planning is required for good demonstration. For this following points should be kept in mind.

- Through the preparation of subject matter.
- Collection of material related to the demonstration.
- Rehearsal of demonstration.

In order to ensure the success of demonstration, the teacher should prepare lesson minutely and very seriously. Depending on the kind of demonstration to be undertaken, pointers or questions may be given to focus students' attention and avoid irrelevant observations.

2. Introducing the lesson

The teacher should motivate students and prepare them mentally for the demonstration. The teacher should introduce the lesson to students keeping in mind the following things.

- individual differences
- Environment
- Experiences

The lesson can also be started with some simple and interesting experiments. Very common event or some story. The experiment should be able to hold the attention of students.

3. Experimentation

The teacher should proceed from whole to parts, by briefly explaining the whole procedure, then to individual steps. Teacher should explain each article in relation to its name, purpose and how to use in the procedure. Make sure that each step is performed in a way, which is understandable by the student and also explain the scientific principles associated with each step.

In between the demonstration procedure, ask questions and encourage the students to seek clarification. The teacher should adapt evaluation technique to ensure that student have learned the skill appropriately.

Steps to carry out effective Demonstrations

The performance in the demonstration table should be ideal for the student

- Carefully plan the demonstration
- Practice the demonstration
- Develop an outline to guide the demonstration
- Make sure everyone can see the demonstration

- Introduce the demonstration to focus attention
- Ask and encourage questions
- Plan a follow up to the demonstration

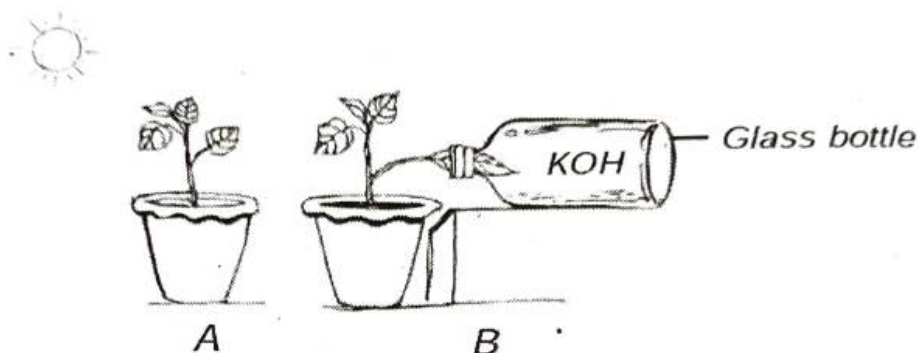
Advantages of Demonstration method

1. It helps in involving various sense to make learning permanent.
2. Teacher gets cooperation of students in teaching learning process.
3. It develops interest in students and motivates them for their active participation.
4. It helps in achieving psychomotor objectives.
5. Even complex skill becomes easy to understand.

Disadvantages

1. It can be used only for skill subjects.
2. Only the attention of the learners are invited towards the activity demonstrated. They are not free to discuss about it.
3. Students are not able to create handmade models for demonstration.
4. It is a time consuming method and the teachers not able to complete the syllabus in time.
5. It is know based on learning by doing.
6. Limited to small group

Demonstration method Example



Both the plants 'A' and 'B' were kept in sunlight after watering. The part of the leaf of plant 'B' which was inserted in the glass bottle containing KOH (Potassium hydroxide) did not turn blue in the iodine test/ starch test, indicating the absence of starch. The part of the leaf outside the bottle turns blue in the said test. Photosynthesis didn't occur in that part of the leaf due to the non-availability of Sun light.

4.1.3. Team Teaching

T - Together

E - Everyone

A - Achieves

M - More

Definition :

All arrangements that include two or more faculty in some level of collaboration in the planning and delivery of a course materials to a class. Team teaching has now found in different departments ranging from school to college level. It can help create a dynamic and interactive learning environment. In ordinary class teaching only one faculty covers entire course material which is interested or uninterested to the students.

But in team teaching course material is divided according to their specialisation and interest. Students have the opportunity to hear multiple perspectives. Different faculties create interest, keep attention and prevent boredom.

Types of Team Teaching

There are different approaches in team teaching. Some of the more common are :

- Interactive team teaching - two faculty members present in front of the class simultaneously.
- Rotational team teaching- faculty members take classes in rotation. This rotational format has a number of variations depending on the subject matter and the number of faculty involved.
- Participant - Observer Team teaching - all participating faculty are present for all the classes but only one is "teaching" at a time. Other teachers could play as participating observer.

Benefits for faculty from Team teaching

1. Improve their own teaching skills
2. Avoid the lonely repetitive fragmented experience of solo teaching.
3. Have opportunities for creative assignments.
4. Develop clearer perspective on the difference between disciplines.
5. Build bridges of understanding across disciplines.

Benefits for students from team teaching

1. Make classes more interesting and challenging because of the novelty.
2. Improve student learning outcomes, interpersonal skills, communication skills, analysis and judgement.
3. Team teaching allows students to involve high level intellectual debate among faculties.
4. Students have the opportunity to see different teaching skills and learning techniques.
5. The presence of another teacher in the class reduces disciplinary problems.

Disadvantages

1. Some teachers are rigid personality type or may carry to a single method of teaching.
2. Some simply dislike the other teachers on the team.
3. Allocation of course material to different faculties may sometimes be inconvenient.
4. Since two or more faculties are present in the class, expenses are more.
5. Teams don't have adequate common planning time.

4.2 Learner-centred methods

In these methods students gather knowledge through discussion, critical thinking, problem solving etc. Teacher's role is to coach and facilitate.

4.2.1 Laboratory method

Instruction must provide experiences and information from which learners can build New knowledge. Instruction helps to focus these processes so that the resulting knowledge is both valid and powerful

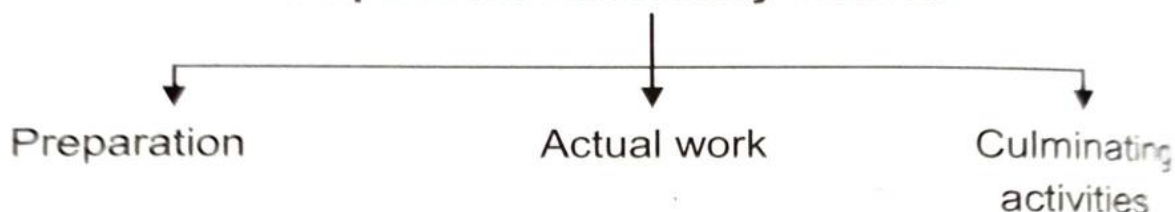
Use of laboratories for Biological Science is primarily concerned with the interaction between students. Laboratory is a place where an individual student interacts with a single Biological. The laboratory is not merely a place that contains machines that students can use. The laboratory is a place where students and teachers interact to learn new skills, to conduct experiments and to broaden their experiences.

Laboratories can be used for both exploration and the illumination of difficult concept. Experimentation should be encouraged at all levels as it is important means for obtaining knowledge. Allow students to explore problems and concepts through the use of established software tools. For instance considerable data collection, tabulating and analysis in a variety of settings may be accomplished with spread sheets - a tool that is readily available. We should encourage the use of such tools in our courses; bring technology in to the classroom.

Objectives of laboratory method

- Better understanding of the subject matter or lesson with the help of practicals.
- Learning by doing
- To make learning easier and more permanent.
- To know the scientific attitudes
- Use the laboratory methods or procedures
- To develop power of observation and reasoning

Steps in the Laboratory method



1. Preparation

- Review of prior knowledge to the present for motivation
- Raising and defining the work to be done
- Expected Learning outcomes
- Planning

2. Actual work

- Students work under the supervision of the teacher.
- Students may work individually or in small groups.
- All works may be recorded for continuity of the task.

3. Culminating Activities

- Reporting what is worked on by individuals or groups
- Conclusions are made.

Advantages of Laboratory Method

1. In this method teacher provides various kinds of learning experiences to the students, as a result of which information gained by them turns out to be of permanent kind.
2. In this method individual difference and interest of all the students are taken into consideration.
3. Through this method students learn to explore various things on their own. They also learn to verify various scientific facts and principles.

4. In this method, an intimate relationship got developed in between the students and teacher.
5. With this method, teacher can develop various good habits among students such as learn to work independently, development of teamwork, honesty, sincerity and truthfulness.

Disadvantages

1. It is a time consuming method.
2. There are some kinds of knowledge which can't be verified through experiments.
3. Due to shortage of resources, we are not able to provide computers and other materials separately for each student.
4. It is very difficult for teachers to provide timely help and guidance to needy students.
5. Backward students are not able to complete the practicals in time.

Laboratory Method - Example - Fermentation

Take a 250 ml beaker filled upto $\frac{3}{4}$ with water. Dissolve 2 table spoon of sugar in it. Add a pinch of yeast powder to the sugar solution. Keep it covered in a warm place for 4 - 5 hours. Now observe the smell of the solution.

Sugar solution is converted to alcohol by yeast and liberation of carbon dioxide is known as fermentation.

4.2.2. Peer tutoring / teaching by students

Peer means students of same category.

Peer tutoring is a student - led instructional strategy used to support. Improved academic achievement. Peer

group members are intimate with each other and so there is no place for shyness or hesitation. Research shows that students involved in peer tutoring show

- higher academic achievement
- Improved relationship with peers
- Improved personal and social development

Peer tutoring programme can be setup to whole class or a group in the class or each one touch one model. High school students are trained to take classes. The following sequence of programme may give better results.

1. Discuss Tutoring with Head of Institution

Before to set up a successful peer tutoring programme, teachers and interested students should discuss the programme with the Headmaster. It is important to make arrangements for a place to tutor, hours when students are available and ensure every student in the class / school is aware of the tutoring programme.

2. Fix the timings

Peer tutoring does not necessarily need to take place after school hours. Students can make arrangements during lunch or during class hours based on the availability of space and needs of students who want tutoring.

3. Set goals

An effective peer tutoring programme needs clear goals and a well organized plan of action to succeed. The goal should focus on simple ideas that students can work toward. Goals for the programme should start with small steps and then work up to a larger.

4. Select Tutors

A successful programme must have students who are motivated and capable of providing help to their classmates. It is important to set clear standards for tutors. They should have material well enough to teach other students. Beyond the knowledge they should have the motivation to help others. Tutor should not have behavioural problems or show any major attendance issues. Class Teacher or subject teacher will select tutors.

5. Provide Training

The training programme focus on teaching the tutors the philosophy of peer tutoring and the techniques that work best. Even if a student have sufficient knowledge it is not always easy to communicate the knowledge to others. Tutors are taught techniques of teaching. Training includes explaining tutors how teaching is often the most effective way of learning, how peer relationships allow students to open up in ways they may not feel comfortable doing with their teachers and how to deal with students who are problem makers.

6. Evaluation

Teachers begin by checking in with their students asking how things are going on. Then teachers spend a couple of minutes reviewing what they did last session.

4.2.3 Project Method

Project is a purposeful activity related to life and it should be carried out in a natural environment in a school or outside the school. Project method is of American origin and is an outcome of Dewey's philosophy of pragmatism. This method is developed and applied practically by Dr. Kilpatrick. In this method student's learn subject matter in natural

setting. Here the content knowledge and skills found in mathematical text book are carried out with the actual happenings in life. Projects can be done individually or in a group or all the students together .

Basic principles of project method

- Learning by doing.
- Learning by living.
- Learning through association, co-operation and activity.

Steps in a project

For completing a project, we have six stages

1. Providing a situation.
2. Choosing a project.
3. Planning of the project.
4. Executing the project.
5. Evaluating the project.
6. Recording all activities of the project.

1) Providing a situation:

A project should arise out of a need felt by students and content of the subject. It should never be forced to them. For this teacher will explain various projects according to need, interest of the students, and their skills to do the project.

2) Choosing a project:

Project will be selected according to the wish of the students. The teacher may give guidance in selection of a better project if the student's make an unwise choice.

3) Planning the project:

The students he encouraged by the teacher to plan out the details of the project. While planning, the following points to be considered. i) the nature and scope of the project. ii) the degree of complexity of project. iii) time allotted to finish the project and iv) availability of material resources. Discussion may be held among students before the final decision is agreed upon.

4) Executing the project:

The teacher helps the student's in assigning work to different student's accordance with their interest, aptitude and capabilities. Each member of the group should be actively involved in the execution of the project. The teacher supervise and guide the students in the execution of the project.

5) Evaluation

The students along with the teacher should review the progress of the project at frequent intervals. The evaluation of the project has to be done in the light of plans, difficulties in the execution and achieved results. Let the students have self criticism and look through their own failings and findings.

6) Record

A complete record of the project be kept by the students. The record should include everything from the beginning to end about the project. It should include the proposed plan, duties allotted to different students and how far were they carried out by them, details of places visited, guidance we received, books consulted and all other details.

Example :

Running of a hostel mess in the school

It involve the following steps.

1. The number of hostlers will be recorded.
2. Secretary of the hostel will be selected by the hostlers.
3. The expected expenditure will be calculated.
4. Expenditure on various heads will be allocated.
5. Budget will be prepared.
6. The amount of collections from the students will be noted.
7. A chart of 'balanced diet' for the hostlers will be prepared.
8. The time of breakfast, lunch, tea and dinner will be notified.
9. Weight of each hostler will be checked after regular intervals and the same will be put on notice board.
10. Punctuality in all the activities will be maintained.
11. Evaluation of all the entire program.
12. Entire activities are recorded and will be informed to the concerned.

Merits of project method

1. It is a psychological method and learner's centered.
2. It introduces democracy in education.
3. It promotes social interaction, inculcates spirit of cooperation and exchanges of experiences among students.
4. It develops self confidence and self-discipline.

5. It makes the learning more interesting and facilitates better understanding of the subject matter as the learning is related to reality.

Demerits

1. The project method is expensive and time consuming and it is not possible to fit into the regular timetable.
2. It does not provide any training in mathematical thinking and reasoning.
3. Systematic and continuous teaching is not possible.
4. The knowledge obtained from various activities are not continuous.
5. All the students are not getting same type of learning.

4.2.4 Individual Activities

An individual activity is primarily authored by students themselves. The individual activities helps students focus on goals and develop future plans and how to use the time in high school to accomplish their personal objectives.

List of Individual Activities

1. Individual Exercises

These provide an opportunity for the students to practice skills. This could include objective type questions and problem solving questions. Exercise should be completed in given time and the teacher verify and give correct answers.

2. Personal design

Given a topic or learning objective, the students are asked to relate it to their real experience.

3. Quiz

Quiz will be conducted for individual students. Answers for the quiz will be also be provided by the teacher.

4. Computer Work

Hundreds of books and topics are available in computer which will open the door of knowledge. Student may be asked to collect points for a topic from computer net work and submit to the teacher.

5. Library work

Students are asked to collect material from the library for a topic given by the teacher.

6. Interview with celebrity

Teacher may invite star celebrity to the class and asks students to prepare 10 questions to ask. Teacher may also asked students to prepare celebrity "profiles".

7. Preparing poster for a function

The teacher may ask his student to design a poster for a function in the school. Student can fill letters with drawings or pictures. After completing the poster it should be hanged on walls or bulletin boards. Students enjoy seeing everyone's design.

4.2.5. Experiential Learning / Experimental Learning

Carl Rogers distinguished two types of learning cognitive and experimental. The former corresponds to academic knowledge and the later refers to applied knowledge. Experimental learning is the process of learning through experience and is more specially defined as learning through "reflection on doing".

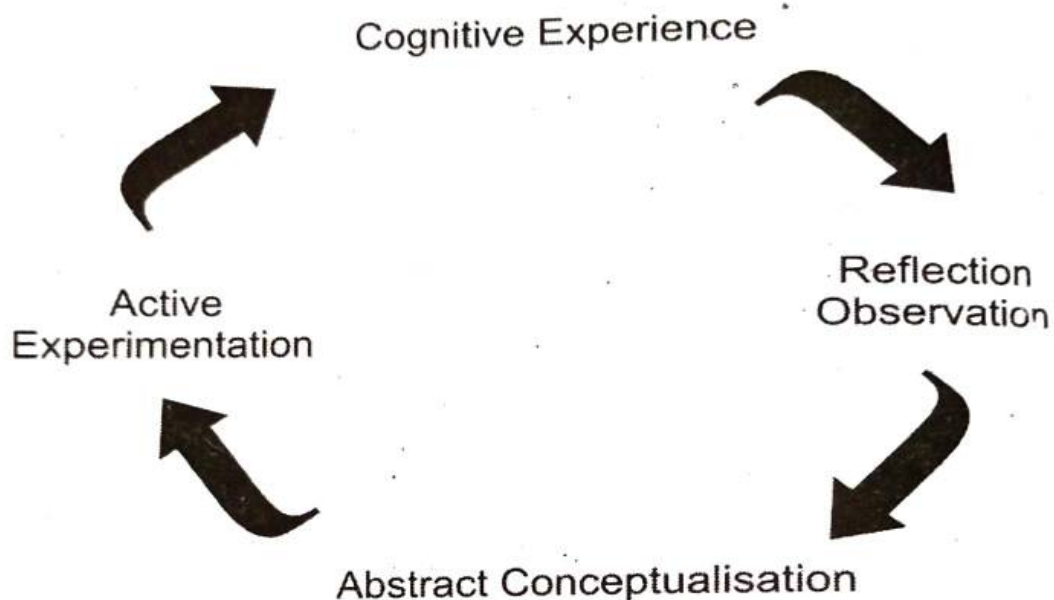
The general concept of learning through experience is ancient. Aristotle wrote, "for the things we have to learn before we can do them, we learn by doing them". In 1970 David A. Kolb helped to develop the modern theory of experimental learning, drawing heavily on the work of John Dewey, Kurt Lewin and Jean Piaget.

David Kolb's Experimental Learning

Experimental learning can exist without a teacher and solely to the meaning - making process of the individual's direct experience. However a genuine learning require certain elements. According to Kolb, knowledge is continuously gained through both personal and environmental experiences. Kolb states that in order to gain genuine knowledge from an experience, the learner must have four abilities.

- The learner must be willing to be actively involved in the experience.
- The learner must be able to reflect on the experience.
- The learner must possess and use analytical skills to conceptualize the experience and
- The learner must possess decision making and problem solving skills in order to use the new ideas gained from the experience.

Four steps of experimental learning



1. **Concrete Experience:** A new experience of situation is encountered or a reinterpretation of existing experience.
2. **Reflective Observation:** Reviewing or reflecting on the experience observation and reflection on that experience.
3. **Abstract conceptualization:** Reflection gives rise to new idea or a modification of an existing abstract concept and leads to generalizations.
4. **Active Experimentation:** The learner used to test hypothesis in future situations, resulting in new experiences.

Effective learning only occurs when a learner is able to execute all four stages of the model.

4.2.6. Teacher - guided learning

Teacher- guided learning helps to develop concepts by themselves and the responsibility for learning rests with

them. This method encourages students to build research skills that can be used throughout their education experiences. This prepare students to lead through a discovery process. Here students are help how to develop questions for this method. It is about students taking control of their learning through a managed process. Students may ask questions like

- How do I solve the problem through this?
- How do I persevere?
- How do I understand the cause and effect relationships that occur in this problem?

"Teachers give students every opportunity to do whatever they want to do in their life". It give them to love learning and to continue on their education.

- Pupils are grouped according to ability or particular learning need.
- The teacher plans the session and support students to begin themselves as learners.
- Students learn to work independently on a particular aspect.

Begin with guided learning

Teacher - guided learning can build background knowledge of the topic before letting students take the control in developing their own learning.

- Teachers start with an overall guiding questions.
- Teachers know what their students to understand beforehand.
- Students know what the outcome of learning will be.

Through teacher guided learning students develop ways to problem solve and prepare them to lead their own learning by the end of the unit. The purpose of this method is to make students being creative problem solvers. For example if students are reading a common text together or posing questions about how they are relating to the text or how they think it might impact the world, there may not be one answer for that. As the teachers start to pose questions, students start to pose those questions for themselves in a way that they can create learning.

Opportunity is given for students to solve their problems in maths. The students are able to share what their thinking was, but then at the same time listen to another child's thinking and recognize there can be different ways to get to the same conclusion or the same outcome and realize that everybody thinks different way and everybody's thoughts are valuable. Parents send their children to school so that they can grow to become resourceful creative contributors to a rapidly changing world.

Teacher - guided learning sequence

1. Focus

Introduction to objective, task and content.

2. Learning check

Identify prior knowledge and confirm the strategies and skills and knowledge to be developed. Teacher initially monitors, identifying strengths and possible areas of attention.

3. Independent work

Pupils work individually or in pairs monitored by the teacher.

4. Rework / Revise

Students are encouraged to return to the task, with support and guidance from the teacher or each other to discuss and revise the work, refining their knowledge understanding and skills. Students strengths are indicated and that pupils are given the confidence and support to revise and reshape their work in a constructive way. This part of the session can produce significant developments and acceleration in learning.

Review:

This session concludes with group reflection on the progress made and strategies used, with the aim of reinforcing the learning objectives, discussing how the skills and knowledge can be applied elsewhere and identifying further targets of learning.

4.2.7. Problem solving method

Problem solving is the process of applying a method not known in advance - to a problem that is subject to a specific set of conditions and that the problem solver has not seen before, in order to obtain a satisfactory solution.

Problem solving method stresses critical thinking and decision making skills. It is a student-centered strategy which require students to become active participants in the learning process.

Essential features of the problem

- The problem should be meaningful, interesting and worthwhile for children.
- It should have some correlation with life.

- It should have some correlation with other subjects if possible.
- It should arise out of the real needs of the student.

Principles for teaching problem solving

- **Select a useful problem - Solving Method**

Problem solving can be difficult and sometime tedious. Show students by your example how to be patient and persistent and how to follow a structured method, such as wood's model.

- **Teach within a specific context**

Teach problem-solving skills in the context in which they will be used. Use real-life problems in explanations and examples.

Help students understand the problem

In order to solve problems, students need to define the end goal. This step is crucial to successful learning of problem - solving skills. If you succeed at helping students answer the questions 'what' and 'why' finding the answer to 'how' will be easier.

- **Take enough time**

When planning a lecture give enough time for : understanding the problem and defining the goal, both individually and as a class; dealing with questions from you and your students; making, finding and fixing mistakes; and solving entire problems in a single session.

- **Ask questions and make suggestions**

Ask students to predict "what would happen if " or explain why something happened. This will help them to develop analytical and deductive thinking skills. Also ask questions and make suggestions about strategies to

encourage students to reflect on the problem-solving strategies that they use.

- **Link errors to misconceptions**

Use errors as evidence of misconceptions not carelessness or random guessing. Make an effort to isolate the misconception and correct it, then teach students to do this by themselves. We can all learn from mistakes.

Wood's problem solving model

1. Define the problem

Identify the problem, list what is known about the problem, and identify the knowledge needed to understand. Once you have list of knowns, identifying the unknowns become easier. Teach students how to select, interpret and use units and symbols. Emphasize the use of units wherever applicable.

All problems have some stated or implied constraints. Teach Students to identify the constraints. Help students to consider from the beginning what a logical type of answer would be and what characteristics will it possess. Create visual images.

2. Think about it

At this stage students will develop a mental image of the problem at hand. Students need to determine by themselves the required background knowledge from illustrations examples and problems covered in the course. Encourage students to collect pertinent information such as conversion factors constants and tables needed to solve the problem.

3. Plan a solution

Type of solution will be determined by the type of problem. Some common problem-solving strategies are;

compute, simplify, use an equation, make a model diagram, table or chart. Help students to choose the best strategy.

4. Carry out the plan

Most problems are not solved quickly or on the first attempt. If a plan does not work immediately, do not let students get discouraged. Encourage them to try a different strategy and keep trying.

5. Look back

Encourage students to reflect. Once a solution has been reached, students should ask themselves the following questions.

- Does the answer make sense
- Does it fit with the criteria established in step (1)
- Did I answer the questions.

What did I learn by doing this?

- Could I have done the problem another way?

Teachers Role in Problem solving

Teacher has a very important role in problem-solving. Some important roles are given below :

1. Give suggestions not answers.
2. Offer a problem solving heuristic method
3. Teach a variety of problem solving strategies
4. Allow time for the students to struggle with the problem
5. Teacher has to give a list of reference books and source books of the problem
6. Provide encouragement and appreciation

Advantages of Problem solving method

1. Students can improve their abilities to retain and recall information.
2. Students learn various methods of problem solving.
3. Students collaborate to solve real-world problems
4. PBL help students apply skills they develop to other real-world scenarios.
5. It helps to improve team work and interpersonal skills.

Disadvantages of Problem solving method

1. PBL activities may not effectively prepare for Tests.
2. Many students may not be prepared to participate in PBL activities.
3. It is a time consuming Assessment.

4.3 Small Group / Whole class interactive learning

In these methods teachers and students solve their ideas through interaction. Teachers and students joined together in their activities. Here group learning is encouraged.

4.3.1. Student Seminar

The word seminar is derived from the Latin word *seminarium* meaning 'seed plot'. A seminar is a form of academic instruction conducted in an educational institution. It has the function of bringing together class students for recurring meetings, focusing each time on some particular subject in which everyone present is requested to actively participate. Here some topics are discussed, questions can be raised and debate can be conducted. The idea behind the seminar system is to familiarize students more extensively with the methodology of their chosen subject and

also to allow them to interact with examples of the practical problems that occur.

Each seminar brings together a group of students to explore and investigate a topic through learning activities including group discussions, research papers and presentations. Students leader or a student nominated by teacher will chair the seminar. Teacher will supervise the functioning of seminar.

Skills needed for presenting seminar

- Introduce the topic at the start of his presentation.
- Reads the presentation aloud.
- Make eye contact with the audience
- Stand and use hand movements to help deliver presentation.
- Indicate what will follow later in his presentation.
- To possible use Information Technology aids for presentation
- Indicate that his presentation is coming to an end
- Invite questions at the end.

Advantages of Seminars

1. Seminars give the wealth of knowledge presented on a particular topic.
2. Seminars are beneficial for those who have difficulty in learning in class room setup.
3. Friendship developed with like interest participants.
4. It helps to become a effective communicator.
5. It helps to collect all points with facts and figures from credible sources in a scientific way.

4.3.2. Group Discussion

Group discussion is a new trend that has come up in order to evaluate student personality. A group of participants are made to discuss on a topic or subject for a limited time and then assessed accordingly.

In group discussion the class is divided into certain homogeneous groups according to their interests, aptitudes, capacities and abilities to do some activities. In group discussion a group of students is assigned a topic for discussion for 15-20 minutes. A teacher or a panel of teachers is looking for an effective combinations of knowledges and skills in the candidates. Knowledge gives the content in a discussion. Without good content you cannot score well. Students need to use their analytical skills to apply theory effectively to solve day to day mathematical problems. Communication skills are most important in group discussion. First train yourself to be a good listener. Develop the patience to listen attentively. Inculcate the good habit of structuring your thoughts and presenting them logically. Writing essays on a variety of topics is good practice in developing thought structure. Learn to respect others for what they are. Learn to be open minded and recognize the fact that people think differently about issues.

Advantages of group discussion

1. It helps to shed away shyness of a candidate and brings his view point amidst all.
2. It stimulates to think in a different new way.
3. It helps the candidate in understanding his own strengths and weakness.
4. It acts as an aid in expansion of the knowledge of the participant.

5. It helps to analyse the content, social or economic issues more logically.

What should be done or avoided in a group discussion

1. Be as natural as possible. Do not try to be someone you are not be yourself.
2. Learn to respect the values and opinions of others.
3. Try to start the group discussion with your opinion if you have complete knowledge of the subject.
4. Do not just go on and on and on your opinion. Complete your ideas in time. Allow others to speak.
5. Use quotes, facts and figures, statements, everyday life examples to express your thoughts.
 - . Think well before you speak.
 - . Make an eye contact with all the participants.
 - . Do not repeat a point or irrelevant point
9. Body language is also important. It says a lot about you.
10. Be an active and dynamic participant.

4.4 Recent trends in teaching:

In these methods students get knowledge through experience or with the help of Audio Visual Aids.

4.4.1. Constructivist learning

Constructivism is basically a theory based on observation and scientific study about how people learn. It says that people construct their own understanding and knowledge of the world, through experiencing things and reflecting on those experiences.

Constructivism is a learning theory found in psychology which explains how people might acquire knowledge and learn. It therefore has direct application to education. The theory suggests that humans construct knowledge and meaning from their experiences.

Constructivism is not a specific pedagogy. Piaget's theory of Constructivist learning has had wide ranging impact on learning theories and teaching methods in education and is an underlying theme of many education reform movements. Research support for constructivist teaching techniques has been mixed, with some research supporting these techniques and other research contradicting those results.

The nature of the learning process

Learning is an active, social process. Social constructivism suggests that knowledge is first constructed in a social context. According to social constructivists, the process of sharing individual perspectives called collaborative elaboration results in learners constructing understanding together that wouldn't be possible alone.

Social constructivist scholars view learning as an active process where learners should learn to discover principles, concepts and facts for themselves, hence the importance of encouraging guesswork and intuitive thinking in learners. In fact, for the social constructivist, reality is not something that we can discover because it does not pre-exist prior to our social invention of it. The reality is constructed by our own activities and that people, together as members of a society, invent the properties of the world.

Characteristics of Constructivist Teaching

Characteristics of a Constructivist Classroom are as follows

- the learners are actively involved
- the environment is democratic
- the activities are interactive and student - centred
- the teachers facilitates a process of learning in which students are encouraged to be responsible and autonomous.

Classroom activities of constructivist learning

In constructivist classroom, students work primarily in groups and learning and knowledge are interactive and dynamic. There is a great focus and emphasis on social and communication skills as well as collaboration and exchange of ideas. Some activities in constructivist classrooms are

Experimentation : Students individually perform an experiment and then come together as a class of discuss results.

Research Projects : Students research a topic and can present their findings to the class.

Field trips : This allows students to put the concept and ideas discussed in class in a real world context. Field trips would often be followed by class discussion.

Films : These provide visual context and thus bring another sense into the learning experience.

Class discussions : The technique is used in all the methods described above. It is one of the most important distinctions of constructivist teaching methods.

Role of Teachers :

In a constructivist classroom, the teacher is perceived as one of the learners who is more experienced and acts like a guide to enable the students to explore some new fields of knowledge. She / he may imply some cognitive strategies to make the lessons more appealing and interesting. The

teacher teaches the student how to make accurate analogies, not to make conclusions or solve different problems. The teacher tries to understand the way learners brain work and she / he leads them to construct and combines the newly - gained knowledge with what students already know from the previous experience.

Method of assessment

In constructivist teaching the process of assessment is based not only on tests but also on observation of student, the students work, way of participation in discussions, students' idea and concept on a topic etc.

4.4.2 Problem-based learning

Problem-based learning (PBL) is a student-centered pedagogy in which students learn about a subject through the experience of solving an open-ended problem. Students learn both thinking strategies and domain knowledge. The goals of PBL are to help students develop flexible knowledge, effective problem solving skills, self-directed learning, effective collaboration skills and intrinsic motivation. Problem-based learning is a style of active learning.

Working in groups, students identify what they already know, what they need to know, and how and where to access new information that may lead to the resolution of the problem. The role of the instructor (known as the tutor in PBL) is to facilitate learning by supporting, guiding, and monitoring the learning process. The tutor must build students' confidence to take on the problem, and encourage the students, while also stretching their understanding. PBL represents a paradigm shift from traditional teaching and learning philosophy, which is more often lecture-based. The constructs for teaching PBL are very different from traditional classroom/lecture teaching.

Meaning

Barrows defines the Problem-Based Learning Model as:

1. Student-centered learning
2. Learning is done in small student groups, ideally 6–10 people
3. Facilitators or tutors guide the students rather than teach
4. A problem forms the basis for the organized focus of the group, and stimulates learning
5. The problem is a vehicle for the development of problem-solving skills. It stimulates the cognitive process.
6. New knowledge is obtained through self-directed learning (SDL).

A well design PBL provides students with the opportunity to develop skills related to

1. Working in teams
2. Managing projects and holding leadership roles
3. Oral and written communication
4. Self awareness and evaluation of group work
5. Working independently
6. Critical thinking and analysis
7. Explaining Concepts
8. Self directed learning
9. Applying course content to real world examples
10. Problem solving across disciplines.

Advantages of problem-based learning

1. It focus student-centred learning
2. Collaborative research work take place instead of Lecturing method
3. It fosters indepth learning
4. It helps in self learning
5. It helps for better understanding of the subject learnt.
6. It helps to work together
7. It create self motivated attitude
8. It enriches the teacher-student relationship

Limitation of Problem-based learning

1. It is a time consuming method.
2. Problem - based learning is new to teachers and they find difficult to change their teaching methods.
3. Teachers find difficult to adopt new assessment method of evaluate the students' achievement.
4. Students are not properly guides by the teachers.
5. All students are not participating in team work.

4.4.3 Brain-based learning

Brain-based learning refers to teaching methods, lesson designs, and school programs that are based on the latest scientific research about how the brain learns, including such factors as cognitive development—how students learn

differently as they age, grow, and mature socially, emotionally, and cognitively.

Brain based learning theory is based on the structure and function of the brain. As long as the brain is not prohibited from fulfilling its normal processes learning will occur. Brain-based education is best understood in three words: engagement, strategies and principles. You must engage your learners and do it with strategies that are based on real science.

The core principle of brain based learning

- Brain can perform several activities simultaneously like tasting and smelling
- Learning engages whole physiology
- The search of meaning is innate
- The search for meaning comes through patterning.
- Emotions are critical of patterning
- The brain processes wholes and parts simultaneously.
- Learning involves both focussed attention peripheral perception.
- Learning involves both conscious and unconscious processes.
- Learning is developmental.
- We have two types of memory spatial and rote
- We understand best when facts are embedded in natural special memory.

Three conditions for learning

Three instructional techniques associated with brain-based learning are

1. Creating learning environments that fully immerse students in an educational experience.

2. Trying to eliminate fear, while maintaining a highly challenging environment.
3. Allowing the learner to consolidate and internalize information by actively processing it.

Impacts on Education

- Curriculum - Teachers must design learning around student interests and make learning contextual.
- Instruction - Teachers let students learn in teams and use peripheral learning. Teachers structure learning around real problems, encouraging students to learn in setting outside the classroom and the school building.
- Assessment - Since all students are learning, their assessment should allow them to understand their own learning styles and preferences. This way students monitor and enhance their own learning process.

Brain-Based Learning suggests the followings

- Teachers must immerse learners in complex, interactive experiences that are both real and rich.
- Students must have a personally meaningful challenge. Such challenges stimulate student's mind to the goal.
- In order for a student to gain insight about a problem, there must be intensive analysis of the different ways to approach it, and about learning in general.
- Feedback is best when it comes from reality, rather than from an authority figure.
- People learn best when solving realistic problems

- Every brain is different, Teachers should allow learners to customize their own environments.
- The best problem solvers are those that laugh.
- Teachers need to realise that the best way to learn is not through lecture, but by participation in realistic environment that let learners new things safely.

4.4.4 Collaborative learning

Collaborative learning is based on the view that knowledge is a social construct. Collaborative activities are most often based on four principles.

- The learner or student is the primary focus of instruction.
- Interaction and "doing" are of primary importance
- Working in groups is an important mode of learning.
- Structured approaches to developing solutions to real-world problems should be incorporated into learning.

Collaborative learning can occur peer-to-peer or in larger groups. Peer learning or peer instruction is a type of collaborative learning that involves students working in pairs or small groups to discuss concepts, or find solutions to problems. This often occurs in a class session after students are introduced to course material through readings or videos before class, and/or through instructor lectures.

Benefits of collaborative learning

Research shows that educational experiences that are active, social, contextual, engaging, and student-owned lead to deeper learning. The benefits of collaborative learning include:

- Development of higher-level thinking, oral communication, self-management, and leadership skills.
- Promotion of student-faculty interaction.

- Increase in student retention, self-esteem, and responsibility.
- Exposure to and an increase in understanding of diverse perspectives.
- Preparation for real life social and employment situations.

What are some examples of collaborative learning or group work activities?

Stump your partner

- Students take a minute to create a challenging question based on the lecture content up to that point.
- Students pose the question to the person sitting next to them.
- To take this activity a step further, ask students to write down their questions and hand them in. These questions can be used to create tests or exams. They can also be reviewed to gauge student understanding.

Think-pair-share/ Write-pair-share

- The instructor poses a question that demands analysis, evaluation, or synthesis.
- Students take a few minutes to think through an appropriate response.
- Students turn to a partner (or small groups) and share their responses. Take this a step further by asking students to find someone who arrived at an answer different from their own and convince their partner to change their mind.
- Student responses are shared within larger teams or with the entire class during a follow-up discussion.

Catch-up

- Stop at a transition point in your lecture.
- Have students turn to a partner or work in small groups to compare notes and ask clarifying questions.
- After a few minutes, open the floor to a few questions.

Fishbowl debate

- Ask students to sit in groups of three.
- Assign roles. For example, the person on left takes one position on a topic for debate, the person on right takes the opposite position, and the person in the middle takes notes and decides which side is the most convincing and provides an argument for his or her choice.

Case study

- Create four to five case studies of similar difficulty.
- Have students work in groups of four or five to work through and analyze their case study.
- Provide 10-15 minutes (or adequate time to work through the cases).
- Walk around and address any questions.
- Call on groups randomly and ask that students share their analysis. Continue until each case study has been addressed.

Review Question

1. Explain any two teacher centred methods of teaching.
2. Explain any two learner centred methods of teaching.
3. Explain any two interactive methods of teaching.
4. Explain any two methods of Recent trends in teaching.